

Build it. Code it. Make it yours.

Teacher guide | Setup, teaching steps, answers and feedback

Grades 6-8 | 8 lessons | 16 classes of 45 minutes

LEGO Education Computer Science & AI Kit 6–8 (#45522, 2026)



This picture shows an idea. Use LEGO's instructions to build the real robot.

Build a rover: a small robot that moves on wheels. Help it carry a paper parcel and stop at a red line. Try it, write down what happens, and make it better.

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Get ready to teach

Who is this course for?

Students in grades 6-8 work in teams of four. They should know how to open a webpage and save a file. They do not need earlier coding experience. Read directions aloud and help with measuring or averages when needed.

Test the real kit first

This is a teaching draft. These coding steps have not yet been tested on a real robot or in a classroom. Before teaching, build each program in Coding Canvas and check it on set #45522. The pictures show ideas, not exact building instructions.

Materials and setup

Use one LEGO set #45522 and one computer or tablet for each team of four. Give each student a job: build, code, start and stop the robot, or write test notes. Swap jobs after each test.

Gather a ruler, tape, flat red and white paper, light cardboard, scissors and tape that is easy to remove. Make a wide lane on a flat floor. Leave lots of empty space beyond the red finish line. Keep the lane away from stairs and table edges.

Before class, build a wheeled base using LEGO's instructions for set #45522. Fit the color sensor so it looks down at the floor, in front of the wheels. A sensor is a part that detects something. Check the real build and take a photo for students to follow. The pictures in this course show ideas; they are not step-by-step LEGO build guides.

Open code.legoeducation.com. This is Coding Canvas, LEGO's coding app. Follow LEGO's setup steps to connect the right team's robot parts. Use the lesson's coding steps to make a program: a list of instructions for the robot. Save it and test it before class. The downloaded steps are a reading guide, not a file the app can run on its own.

Charge the robot parts as LEGO directs. Lift the wheels off the floor for the first check. Test which way they turn, how to stop the program, and how to turn off the robot. Keep the computer or tablet connected while the robot runs.

Before students arrive

Check each team's robot

Charge the parts. Connect the correct robot to the correct device. Open the saved code. Raise the wheels and check which way they turn. Test the screen Stop control and how to turn the power off. With the wheels still, check red, white and another color. Then try one slow drive that stops at red.

Arrange the room

Mark a separate lane for each team. Leave extra space beyond each finish line. Label the kits. If possible, keep one working robot ready to help find problems with another team's setup. Keep each robot's test notes with its kit.

Keep students safe

Ask the teacher before each run. Start slowly. Keep one person ready at the Stop control. If the robot moves the wrong way, stop at once. If the screen control does not work, use the power-off method the teacher showed you. Test both ways to stop before students drive.

Keep fingers, hair and loose clothes away from the wheels. Turn the robot off before changing its parts. Stop if the sensor cannot name the color or the connection is lost. A red floor line tells this robot where to stop. It cannot use that color sensor to spot people, walls or table edges.

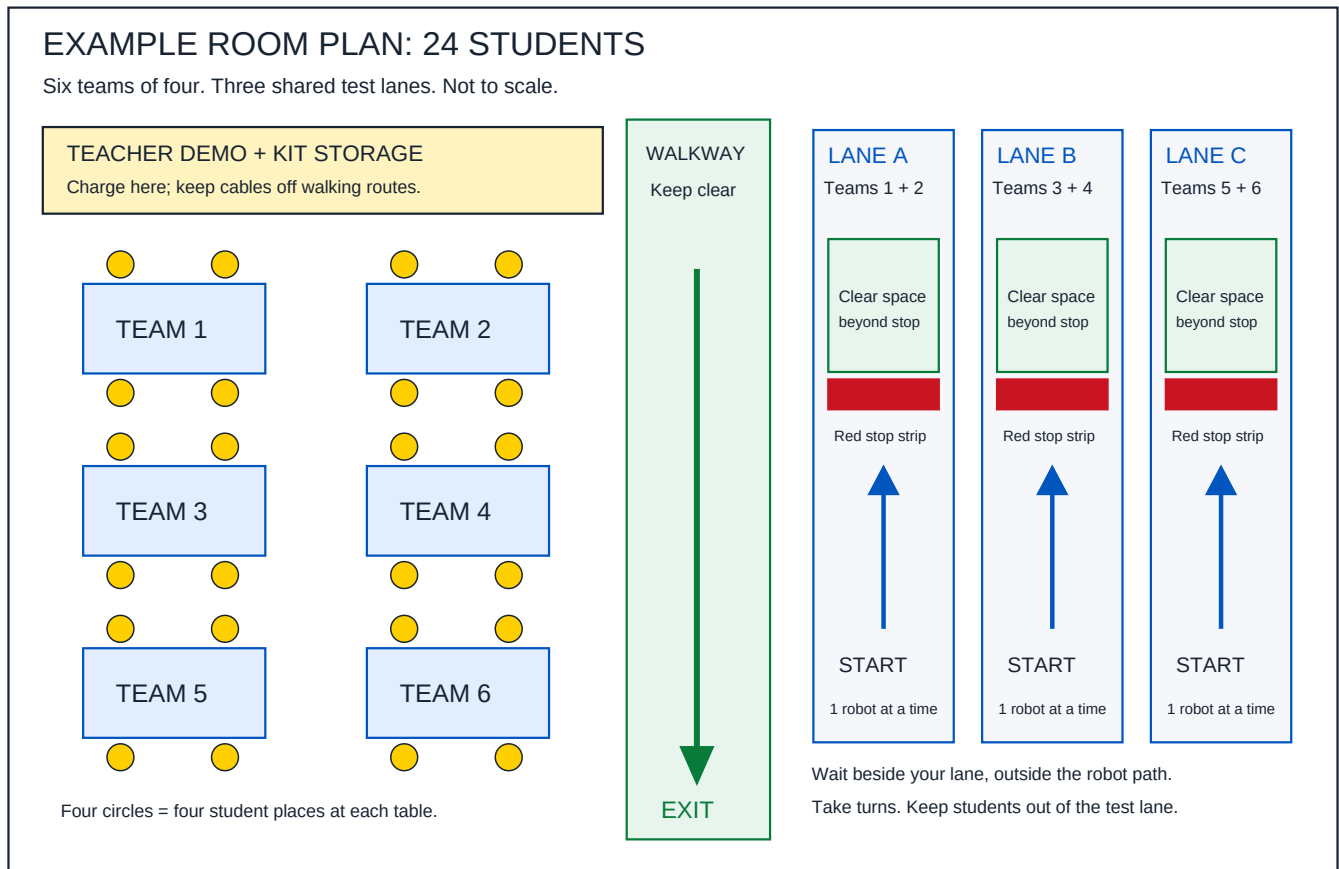
Two classes per lesson

First class: share the goal and make a prediction (5 minutes); show the first steps (10); build and code (20); try one test (5); save and tidy (5). A prediction is what you think will happen and why.

Second class: review (5 minutes); test and improve (25); explain your results to another team (10); answer one short question, save and tidy (5).

Set up the classroom

An example for six teams of four. Move the tables and lanes to fit your room. Keep a clear route to the real exit.



Before testing

Make every lane at least twice as wide as the robot. Leave clear floor beyond each stop strip. Decide the actual lane length using the teacher's slow-drive test. The picture gives zones, not room measurements.

Who uses each lane?

Teams 1 and 2 share lane A. Teams 3 and 4 share lane B. Teams 5 and 6 share lane C. Only one robot uses each lane at a time. The other team builds, writes notes or waits outside the lane.

Where do students stand?

The test leader stands beside the lane at the Stop control. Observers stay outside the taped boundary. Turn power off before carrying the robot back. Keep charging cables on the storage table.

In the online Teacher Desk, hover over, focus on or tap "See an example classroom setup" to open this plan.

Help everyone take part

Show it, say it, try it

Read the new words aloud and point to the part or demonstrate the action. Show one step, then give students time to try. Offer the audio and its written words. Keep a worked example on the display.

Match support to the learner

Follow each student's documented support plan with the school's support staff. Offer extra time, larger print, a reachable work area, or an oral or drawn response when appropriate. Keep the learning goal clear. Check each student's understanding, even when the team shares one robot.

For students learning English

Pair each key word with a picture or real part. Let students discuss an idea in their home language before sharing in English. Model a short answer and give time to rehearse it with a partner. These packets are currently in English.

Give students a starting sentence

"I predict ____ because ____."

"We changed ____ and kept ____ the same."

"Our test showed ____."

"Next we will ____ because ____."

Share the jobs

Rotate building, coding, starting and stopping, and recording after each test. Make tools and Stop controls easy to reach. A student who needs help writing can explain a result while a partner records it.

Adjust the challenge

For extra support, use the teacher-prepared program and change one message or setting. For an extension, change one design constraint, such as parcel size, and explain how to make the comparison fair. Speed is not the goal.

Without a kit

Use paper models and the Mission simulator to practice planning. Label results "paper model" or "screen test." Complete the movement and color tests later on the physical robot.

Course goals and evidence

By the end, students should be able to explain and demonstrate the skills below. Evidence means work or an explanation that shows what they learned.

Students can...	Practice	Evidence to collect
Explain how a color input leads to an output.	Modules 1, 4 and 6: test colors and apply a rule.	Color-test notes; individual explanation; slide exit checks.
Put short movements in order and explain how a run stops.	Modules 2, 5 and 6: measure a drive, plan a route and add a stop rule.	Saved program, route plan and a safe demonstration.
Compare designs using fair tests.	Modules 3, 4 and 7: change one thing and repeat tests.	Before/after results and a reason for the chosen change.
Improve and explain a robot that meets agreed rules.	Module 8: choose one build, test and revise it.	Chosen-build packet, saved code, directions and individual explanation.

Check before and after

Before module 1, ask: “What does a sensor do?” “How could a robot choose an action?” and “What makes a test fair?” Accept a guess or “not yet”; use the answers to plan support. Ask the same questions after module 8 and have students point to a result from their own work.

Feedback and optional grades

This course does not assign a required grade or pass mark. Use the rubric for feedback. If your school requires grades, explain your local scoring policy before the project. Collect an individual explanation as well as team work. Simulator stars are not grades.

The eight lessons

Lesson	What students make or do	Main idea
1. Wake up your rover	Make the screen respond to a color	A color goes in; a message comes out
2. Make a short drive	Drive a little, then stop	Short drives and measuring
3. Build a parcel holder	Keep a paper parcel on the robot	Build and compare two holders
4. Help your robot read colors	Find a finish line the sensor can read	Check color readings
5. Plan the delivery route	Drive, turn and reach the delivery space	Put steps in order
6. Stop at the red line	Use a color reading to choose when to stop	Use color to choose move or stop
7. Make the delivery	Carry the parcel and stop in the right place	Test three rules for success
8. Choose a build and show it	Give your robot a new job	Explain changes and results

Plan the time

Use two classes a week for eight weeks, or four classes a week for four weeks. Each class lasts 45 minutes. Keep time for repeat tests and explanations. Add a setup class if the robots are not built yet.

Wake up your rover

What students should learn

Show a color to the sensor and explain the message that appears.

Words to know

Sensor: A part that detects something, such as a color.

Program: The steps the robot follows.

Input / output: The color reading goes in. A screen message comes out.

First class: show, build and try

Use the matching workbook page. Read the new words and point to the parts. Ask “What do you think will happen, and why?” Show the first step. Let teams build and try one test. Before driving, check the lane and make sure someone is ready to stop the robot.

Second class: test and improve

Have teams repeat the test and write down every result. Change one thing and try again. Ask “What stayed the same?” and “What did your notes show?” Finish with the question below. Save the code.

What to look for

Each of the three colors gives the right message. The wheels do not move.

Question and example answer

What is the input in this test? What is the output?

The input is the color reading. The output is the screen message.

Common mix-up

The robot does not know what DELIVERY means. It follows the rule you wrote.

If it does not work

Check that you connected the right robot. Make sure the sensor can see the paper. If a wheel moves, stop and ask the teacher to check the code.

Try more

Make messages that use words or symbols, so students do not have to tell screen colors apart.

Make a short drive

What students should learn

Make the robot stop after a short time. Measure how far it travels.

Words to know

Motor: A part that makes something move.

Bounded: Set to stop after a certain time or number of steps.

Trial: One test run.

First class: show, build and try

Use the matching workbook page. Read the new words and point to the parts. Ask “What do you think will happen, and why?” Show the first step. Let teams build and try one test. Before driving, check the lane and make sure someone is ready to stop the robot.

Second class: test and improve

Have teams repeat the test and write down every result. Change one thing and try again. Ask “What stayed the same?” and “What did your notes show?” Finish with the question below. Save the code.

What to look for

The robot stops after each of three short drives. You also show that the Stop control works.

Question and example answer

Why might the same code move the robot a different distance tomorrow?

The floor, battery charge, wheels or load may be different. Measure the distance each time instead of guessing.

Common mix-up

A speed setting of 20 does not mean 20 centimetres per second.

If it does not work

Stop. Check the charge, connection and wheel space. Test the wheel direction with the wheels raised before trying again.

Try more

Try a second low speed. Do three runs at each speed. Compare the distances.

Build a parcel holder

What students should learn

Build two versions of a holder and use your tests to choose one.

Words to know

Cargo: The thing your robot carries.

Cradle: A holder that keeps the parcel in place.

Fair test: Change one thing and keep the rest the same.

First class: show, build and try

Use the matching workbook page. Read the new words and point to the parts. Ask “What do you think will happen, and why?” Show the first step. Let teams build and try one test. Before driving, check the lane and make sure someone is ready to stop the robot.

Second class: test and improve

Have teams repeat the test and write down every result. Change one thing and try again. Ask “What stayed the same?” and “What did your notes show?” Finish with the question below. Save the code.

What to look for

Your holder keeps the parcel on in at least two of three runs. It does not rub a wheel or cover the sensor.

Question and example answer

Why should you keep the same parcel when you compare two holders?

A different parcel might change the result. Keeping it the same helps you test the holder itself.

Common mix-up

A taller holder is not always better. It can make the robot tip or block a part.

If it does not work

Turn the power off. Check for rubbing, loose tape or a top-heavy load. Change one part, then test again.

Try more

Give your drawing to another team. Can they make the same holder without extra help?

Help your robot read colors

What students should learn

Test paper colors and choose a setup that gives clear readings.

Words to know

Reading: The information a sensor gives you.

Unknown reading: The sensor cannot name the color.

Fair test: Change one thing at a time.

First class: show, build and try

Use the matching workbook page. Read the new words and point to the parts. Ask “What do you think will happen, and why?” Show the first step. Let teams build and try one test. Before driving, check the lane and make sure someone is ready to stop the robot.

Second class: test and improve

Have teams repeat the test and write down every result. Change one thing and try again. Ask “What stayed the same?” and “What did your notes show?” Finish with the question below. Save the code.

What to look for

You record nine readings, try one change and explain which paper setup you will use.

Question and example answer

The sensor names red correctly only twice in three tries. What should you do before driving?

Keep the wheels still. Check the light, paper and sensor height. Change one thing and test again.

Common mix-up

A color sensor reads colors. It does not tell you how far away a wall is.

If it does not work

Hold the paper flat. Keep the sensor steady. Reduce glare, or try another sheet of paper. Change one thing at a time.

Try more

Count how many readings were right. Would you trust the setup after only three tries? Explain why.

Plan the delivery route

What students should learn

Put steps in order, then test and adjust a short route.

Words to know

Sequence: Steps in order.

Calibrate: Test and adjust the settings.

Variable: One thing you can change, such as drive time.

First class: show, build and try

Use the matching workbook page. Read the new words and point to the parts. Ask “What do you think will happen, and why?” Show the first step. Let teams build and try one test. Before driving, check the lane and make sure someone is ready to stop the robot.

Second class: test and improve

Have teams repeat the test and write down every result. Change one thing and try again. Ask “What stayed the same?” and “What did your notes show?” Finish with the question below. Save the code.

What to look for

The robot stops inside your marked space in at least two of three runs.

Question and example answer

Why does one good run not prove the route will work every time?

Small changes in the floor, charge or start position can change the result. More runs help you spot a pattern.

Common mix-up

A perfect turn on a screen does not set the correct turn time for your real robot.

If it does not work

Test the straight parts and turn on their own. Check the starting direction before changing the code.

Try more

Compare two routes to the same space. Which takes fewer steps? Which works more often?

Stop at the red line

What students should learn

Make the robot move on white and stop on red or an unknown reading.

Words to know

If-then rule: If this happens, do that.

Loop: Steps that repeat.

Bounded: Set to stop after a certain time or number of steps.

First class: show, build and try

Use the matching workbook page. Read the new words and point to the parts. Ask “What do you think will happen, and why?” Show the first step. Let teams build and try one test. Before driving, check the lane and make sure someone is ready to stop the robot.

Second class: test and improve

Have teams repeat the test and write down every result. Change one thing and try again. Ask “What stayed the same?” and “What did your notes show?” Finish with the question below. Save the code.

What to look for

Three runs stop when the sensor reads red. Unknown readings also stop the robot. It never restarts on its own.

Question and example answer

Why might the robot miss a narrow red strip even though it reads red while still?

It moves between readings. It may pass the strip before the next reading. A wider strip or lower speed gives it more time to see red.

Common mix-up

This rule looks for a floor color. It cannot spot a person or wall in the way.

If it does not work

Stop driving. Check the colors while still, then check the strip width and code order. Ask the teacher before another floor run.

Try more

Show a message explaining the stop: RED LINE, CHECK COLOR or TIME UP.

Make the delivery

What students should learn

Join your tested parts and check the whole delivery three times.

Words to know

Success criteria: The things that must happen for a test to count as a success.

Version: One saved design or program before you change it.

Reliable: Works as expected again and again.

First class: show, build and try

Use the matching workbook page. Read the new words and point to the parts. Ask “What do you think will happen, and why?” Show the first step. Let teams build and try one test. Before driving, check the lane and make sure someone is ready to stop the robot.

Second class: test and improve

Have teams repeat the test and write down every result. Change one thing and try again. Ask “What stayed the same?” and “What did your notes show?” Finish with the question below. Save the code.

What to look for

At least two of three runs meet all three rules. Keep a record of every run, including the ones that fail.

Question and example answer

Your first version passed once in three runs. Your new version passed twice. What can you say?

The new version did better in these tests. You need more tests before saying it works every time.

Common mix-up

Reaching the finish does not count as a full delivery if the parcel falls off.

If it does not work

Check the holder, movement and color reading one at a time. Then try them together again.

Try more

Add a new delivery rule. Explain what you would need to change and how you would test it.

Choose a build and show it

What students should learn

Choose one of five builds, test it and explain how you improved it.

Words to know

Attachment: A part you add to the robot for a new job.

Evidence: Measurements and notes that show what happened.

Handover: Directions that help someone else use your robot.

First class: show, build and try

Use the matching workbook page. Read the new words and point to the parts. Ask “What do you think will happen, and why?” Show the first step. Let teams build and try one test. Before driving, check the lane and make sure someone is ready to stop the robot.

Second class: test and improve

Have teams repeat the test and write down every result. Change one thing and try again. Ask “What stayed the same?” and “What did your notes show?” Finish with the question below. Save the code.

What to look for

The chosen build meets its own rules in at least two of three runs. Another team can follow your directions. Explain one improvement and one remaining problem.

Question and example answer

What should you do if your robot fails during the class demonstration?

Stop it. Explain what happened. Show your earlier results and name one thing you would test next. Do not hide the failed run.

Common mix-up

A good project does not need a perfect last run. Clear, honest test notes matter too.

If it does not work

Use your saved code and setup drawing to check what changed. If needed, show one part that works and explain what still needs fixing.

Try more

Plan a different build choice for a later lesson. Explain which parts and tests would need to change.

Score the learning

A rubric is a scoring guide. Give 1 to 4 points in each row, for up to 16 points. Ask each student to explain their part. If a student needs a safety reminder, stop and teach that step again before another drive.

What to check	4: Can explain and do it	3: Mostly there	2: Needs some help	1: Needs a new start
Goal and setup	Clear rules for success. Another team can follow the setup.	Clear rules. Setup needs one small detail.	Some rules or setup steps are missing.	Cannot yet explain the goal or setup.
Explain the code	Explains the steps, choices and what the code cannot do.	Explains the steps and how the robot stops.	Explains some steps with help.	Cannot yet link the code to the action.
Test and improve	Repeats fair tests. Uses results to explain a change and what needs more work.	Repeats tests and uses the notes to choose a change.	Some tests or notes are missing, or too many things change at once.	Needs help to test and keep useful notes.
Share the work	Another team can follow the directions. Each student explains their part.	Directions are clear. Students share jobs.	Directions need detail, or jobs are not shared well yet.	Directions are hard to follow. Individual work is not clear yet.

Example feedback

“You explained your stop rule clearly. You changed both speed and marker width in your tests. We cannot tell which change helped. Keep the speed the same. Test two marker widths three times each.”

Optional extra: sort colors on the screen

Keep the wheels still. Try two colored parcel labels and a new color. Make a table that links each color reading to a screen message. Explain what happens when the color is unknown. A robot that physically sorts parcels would need a separate build and new tests.

Use the Mission simulator

The screen robot teaches instructions in order and choosing an action. A forward step moves one square. A turn is a quarter-turn.

Teach one run

Choose a map. Have students predict the finish, add instructions and use “One step” to explain each action. After a run, save the test result. Compare a failed attempt with a revised route. Download test notes before refreshing or closing the page.

Explain “If blocked”

It checks the next square. If a wall or the grid edge is ahead, the robot turns right. Otherwise, it moves one square forward. It makes one choice each time the instruction runs; it does not find a whole route by itself.

Example routes

F = forward one square. R = turn right. I = If blocked.

First delivery: F F F R F F F F

Around the corner: F R F F F F R F

Sensor decision: F I F F F F

Use the game feedback

A bump is a reason to inspect the next square and revise the plan. Celebrate useful explanations and improvements. Stars and delivery graphics are encouragement, not assessment scores. Use the sound control to mute effects when needed. The game respects the device’s reduced-motion setting.

Connect screen practice to the real kit

The screen robot checks walls. The physical robot in these lessons reads floor colors. Ask: “What information goes in, what rule is used, and what action comes out?” Real wheels can slip and color readings can vary. Screen success does not prove the physical build works.

Print for each class

Open the course's "Print student packets" page. Choose the module and class; for module 8, choose the team's final build. The selected packet includes the pages needed for that class.

Which packet?	Pages and paper
Modules 1–7, class 1	4 pages / 2 sheets
Modules 1–7, class 2	2 pages / 1 sheet
Module 8, chosen build, either class	4 pages / 2 sheets

Printer settings

Use US Letter, Actual size (100%), two-sided, flip on long edge. Each class packet begins with Name, Date and Period. Print the classroom pack's templates at actual size and check the scale bar with a ruler.

What the packets include

Build steps, labeled diagrams, coding steps, prediction prompts, test tables and reflection space. The complete module PDF is a reference copy. The full student workbook also includes word help and the dated kit-cost sheet.

Before handing them out

Prepare the checked robot base and the physical supplies. Open the matching lesson slides and coding guide. For modules 1 and 4, keep the wheels still: the program checks colors 60 times with a 0.5-second pause. This gives about 30 seconds; run it again if students need more time.

Collect evidence of learning

Collect the class packet and check one explanation from each student. Keep the team's saved code and test notes together. The slide answer key at the back helps you discuss the screen activities.

Teach the five final choices

Choice	Teacher check	Program to check
Parcel delivery	Low tray; light parcel; sensor clear	Lesson 6 color stop; max 20 checks
Drawing robot	Washable marker; protected floor; tip does not lift wheels	Short timed forward drive, then Stop
Paper sweeper	Soft paper only; bristles bend; wheels stay clear	Short timed forward drive, then Stop
Ball pusher	Light paper ball; wide clear lane; V not a clamp	Short timed forward drive, then Stop
Rescue carrier	Paper figure only; wide tow bar away from wheels; straight route	Short timed forward drive, then Stop

Score all five choices fairly

Use the same four-row scoring guide. Judge each build against its own three success rules. Students should record three runs, use a result to explain one change and give clear start-and-stop directions. A sweeper or ball pusher is supposed to touch its paper test object; it should not touch people or unrelated objects.

Keep the explanations accurate

Only the delivery choice uses the red-line stopping program. The other four start with a short timed drive. A marker holder draws; it does not read the drawing. A V guide pushes a ball; it does not grab it. The rescue carrier tows a paper figure; it cannot rescue a person.

Help without doing the thinking

Ask: “What should your robot do?” “Which rule did this run miss?” “What one thing could you change?” If a build struggles, stop and check weight, rubbing and attachments before changing the code.

Slide answers | 1–2

Use these answers to discuss reasoning. Let students revise an answer after feedback. Screen checks do not replace the physical build or an individual explanation.

Module 1: Wake up your rover

Prediction: The sensor reads red. What should happen?

Answer: DELIVERY appears. The wheels stay still.

Yes. Red changes the message to DELIVERY.

Practice: Put these steps in order. Use the arrows to move them.

Answer: Show red paper to the sensor → Use the rule for red → Show DELIVERY on the screen

First show the paper. The program uses its rule, then shows the message.

Exit check: How can the robot choose a message without a person choosing it each time?

Answer: Use the color reading and a rule in the program.

Yes. The program uses the color reading to choose.

Module 2: Make a short drive

Prediction: You run the same code tomorrow. Must the robot travel exactly the same distance?

Answer: No. The floor or battery charge may be different.

Yes. Test and measure each run.

Practice: Three test runs travel 18 cm, 20 cm and 22 cm. What is the mean distance? Mean means average: add the distances, then divide by the number of runs.

Answer: 20 cm

The average is 20 cm. The next run may travel a different distance.

Exit check: You want to test a new speed. What should stay the same?

Answer: Drive time, floor, parcel and starting direction

Yes. Change only speed so you can see what it changes.

Slide answers | 3–4

Use these answers to discuss reasoning. Let students revise an answer after feedback. Screen checks do not replace the physical build or an individual explanation.

Module 3: Build a parcel holder

Prediction: Your parcel falls out. What should you try next?

Answer: Change one part of the holder and repeat the same drive.

Yes. Compare the two holders using the same test.

Practice: Choose TWO safe ideas for holding the parcel.

Answer: A low, light holder that leaves room for wheels and sensor; A holder you can remove, with Stop controls easy to reach

Keep the holder light and low. Leave the wheels, sensor and Stop controls clear. Ask the teacher to check it.

Exit check: Why use the same parcel to test two holders?

Answer: So the holder is the only thing you change

Yes. A different parcel might change the result too.

Module 4: Help your robot read colors

Prediction: The sensor reads red correctly two times out of three. What next?

Answer: Keep the wheels still, check the setup and test again.

Yes. Try changing one thing, such as the sensor height.

Practice: You test a color nine times. Six readings are correct. How many are wrong or unknown?

Answer: 3 readings

$9 - 6 = 3$. Keep those three readings in your notes too.

Exit check: How can you test whether the light changes the reading?

Answer: Change the light. Keep the paper and sensor height the same.

Yes. Change just the light.

Slide answers | 5–6

Use these answers to discuss reasoning. Let students revise an answer after feedback. Screen checks do not replace the physical build or an individual explanation.

Module 5: Plan the delivery route

Prediction: The screen robot faces north (up). It turns right, then moves forward. Which way does it go?

Answer: East (right)

Yes. Turning right from north points east.

Practice: Start facing north (up). Go two squares north, then three east (right), and stop. Put the steps in order.

Answer: Go forward two squares → Turn right → Go forward three squares → Stop

Go north, turn right, go east, then stop. Test the real robot's drive times separately.

Exit check: Your real robot makes the right turn once. What do you know?

Answer: It made the right turn in that one test.

Yes. Try again and write down all the results.

Module 6: Stop at the red line

Prediction: The sensor reads red as the robot drives toward the finish. What next?

Answer: Stop and end this test.

Yes. Keep it stopped until you start a new test.

Practice: Choose white, red and unknown, one at a time. Guess what should happen, then test each reading.

Answer: White: a short move. Red or unknown: stop and end the test.

Only white allows a short move. Red or unknown ends the test.

Exit check: The robot has stopped on red. It then sees white. What should happen in this same test?

Answer: It stays stopped.

Yes. A person must start a new test.

Slide answers | 7–8

Use these answers to discuss reasoning. Let students revise an answer after feedback. Screen checks do not replace the physical build or an individual explanation.

Module 7: Make the delivery

Prediction: The robot stops in the space and hits nothing, but drops its parcel. Does the run pass?

Answer: No. All three rules must be met.

Yes. All three things must happen.

Practice: How many test runs below meet all three rules?

Answer: 1 passing run(s)

Only run 1 has three Yes answers. The goal is at least two passing runs out of three.

Exit check: Your first holder passed 1 of 3 runs. Your new holder passed 2 of 3. What can you say?

Answer: It worked better in these tests.

Yes. Say how many tests you did and what happened.

Module 8: Choose a build and show it

Prediction: Your robot does not finish during your presentation. What should you do?

Answer: Stop, explain what happened and name one next test.

Yes. A clear explanation helps people understand your work.

Practice: Put these parts of a two-minute presentation in order.

Answer: Explain the robot's job and the three rules for success → Explain the steps in your robot's program → Show test results before and after your change → Say what needs more work and what you would test next

Start with the job. Explain how it works. Show your results. Finish with what you would try next.

Exit check: You have only tried the screen robot. Which statement is true?

Answer: I still need to build and test the real robot.

Yes. Test the real build with your teacher.

Sources and teaching resources

Use the official resources below when preparing the physical kit. The lesson guides give written coding steps, not app project files. Make, save and test the programs in Coding Canvas before teaching; block names may differ.

Selected kit and current US pricing

<https://education.lego.com/en-us/products/lego-education-computer-science-and-ai/45522/>

Official #45522 build instructions

<https://www.lego.com/service/buildinginstructions/45522>

Coding Canvas

<https://code.legoeducation.com/>

Teacher Portal and LEGO lessons

<https://teach.legoeducation.com/>

LEGO Education Python reference (optional advanced route)

<https://github.com/LEGO/LEGOEducation>

SPIKE retirement and support

<https://education.lego.com/en-us/spike-update-2026/>

Original Rise course

<https://luvbuniz.github.io/robotics/>

Official starting model C103: pages 24-31

<https://www.lego.com/cdn/product-assets/product.bi.core.pdf/6640383.pdf#page=24>

Credits

Course design and writing: Amy Sullivan. Audio uses Amy's own voice clone. AI-made illustrations show project ideas, not verified LEGO builds. This independent course is not endorsed by the LEGO Group.

Readiness

This teaching draft still needs physical-kit and classroom trials. Screen checks and document review do not verify hardware performance. Kit details and US planning prices were checked September 10, 2026; confirm prices before ordering.

Word help

A glossary is a list of words and their meanings. Use this page when you meet a word you do not know.

Word	Plain English meaning
Rover	A small robot that moves on wheels.
Program	A list of steps a computer or robot follows.
Coding block	A piece on the screen that gives the robot an instruction.
Sensor	A part that detects something, such as a color.
Motor	A part that uses electricity to make something move.
Input / output	Input is information going in. Output is the action or message that comes out.
Bounded	Set to stop after a certain time or number of steps.
Trial	One test run.
Fair test	Change one thing. Keep the other things the same.
Cargo / cradle	Cargo is what the robot carries. A cradle is a holder that keeps it in place.
Calibrate	Test and adjust settings so the robot does what you need.
Sequence / loop	A sequence is steps in order. A loop repeats steps.
Variable	One thing that can change, such as speed or time.
Mean	An average: add the numbers, then divide by how many numbers you added.
Success criteria	The things that must happen for a test to count as a success.
Evidence	Measurements or notes that show what happened.
Attachment	A part you add to the robot for a new job.

What will the kits cost?

These are US prices for LEGO Education Computer Science & AI 6-8 (#45522). The supply amounts are estimates: money set aside for materials.

What to buy	US dollars	What the amount covers
One kit · up to 4 learners	\$529.95	LEGO's listed price
Six individual kits · 24 learners	\$3,179.70	6 × \$529.95; fewer charging extras than bundle
Classroom bundle · 24 learners	\$3,499.00	6 kits plus charging and replacement extras
Supplies for a small tryout	\$35.00	Money set aside for: paper, tape, ruler and cargo materials
Supplies for six teams	\$120.00	Money set aside for six teams
Small tryout total before tax / shipping	\$564.95	1 kit + \$35 supplies; using a device you already have
Class total before tax / shipping	\$3,619.00	1 bundle + \$120 supplies; using 6 devices you already have

Prices were checked on LEGO's US website. Check them again before buying. The totals do not include tax, shipping, computers, school computer support or teacher training. The course and screen practice area do not need a paid subscription. No kit has been bought.

What comes with the class bundle?

The bundle has six kits, three extra multi-chargers, eight extra connection cards and a replacement brick pack. Buying six single kits costs \$319.30 less, but does not include all those extras.

Start small

Try one kit with four students and a computer or tablet you already have. For 24 students in teams of four, use six kits. For pairs, 12 kits cost \$6,359.40 before other costs. One spare kit adds \$529.95. These are kit purchases, not monthly fees.

Buying an older kit?

EV3 and Robot Inventor use different parts and coding apps. These lessons' coding steps would need changes. SPIKE direct sales ended June 30, 2026; LEGO says its app will be supported through June 30, 2031. Before buying used, check that the parts, charger and app you need are available.

[Selected kit and current US pricing](#)

[SPIKE retirement and support](#)